

KELSEY DAM, MB, CanadaWMO: **715720**Lat: **56.038N**Lon: **96.510W**Elev: **600**StdP: **14.38**Time Zone: **-6.00 (NAC)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.0	-31.4	-42.8	0.5	-35.7	-37.9	0.6	-31.3	18.0	-3.5	16.2	-5.4	3.4	250	0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	81.1	62.7	78.3	61.4	75.2	60.3	66.3	76.0	64.2	73.5	62.3	71.7	8.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.0	88.2	69.9	60.5	80.6	67.8	58.5	74.9	66.0	31.4	75.9	29.7	73.3	28.3	72.0	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.7	14.0	DB	-42.8	88.1	4.8	3.8	-46.3	90.9	-49.1	93.1	-51.9	95.2	-55.4	98.0	(4)
(5)			WB	-42.9	69.8	4.6	2.5	-46.2	71.6	-48.8	73.0	-51.4	74.4	-54.7	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	-8.0	-5.9	8.3	26.6	41.6	54.6	62.0	58.0	48.1	33.6	12.8	-5.1	(6)	
	DBStd	27.37	13.20	11.92	14.05	13.51	9.87	8.02	6.26	6.53	8.49	8.27	13.10	14.54	(7)	
	HDD50	9184	1799	1567	1293	703	297	42	2	10	137	510	1116	1708	(8)	
	HDD65	13797	2264	1987	1758	1151	727	322	133	233	511	972	1566	2173	(9)	
	CDD50	930	0	0	0	2	36	180	373	257	79	3	0	0	(10)	
	CDD65	69	0	0	0	0	1	10	40	15	3	0	0	0	(11)	
	CDH74	893	0	0	0	0	29	208	454	172	30	0	0	0	(12)	
	CDH80	142	0	0	0	0	6	31	79	22	4	0	0	0	(13)	
	WSAvg	6.8	6.1	6.3	6.8	7.3	8.1	7.4	6.4	6.5	6.8	7.3	6.5	5.7	(14)	
	PrecAvg	19.3	0.7	0.6	0.8	0.9	1.6	2.4	3.2	3.1	2.3	1.9	1.1	0.7	(15)	
(16) Precipitation	PrecMax	26.9	1.5	1.3	2.3	2.0	3.2	4.1	5.4	7.8	3.9	4.1	2.6	1.3	(16)	
	PrecMin	14.8	0.3	0.1	0.2	0.3	0.5	0.6	1.7	1.3	1.1	0.8	0.6	0.3	(17)	
	PrecStd	2.6	0.3	0.3	0.5	0.4	0.6	0.7	1.1	1.5	0.9	0.8	0.4	0.3	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	29.9	47.3	65.9	77.7	83.5	86.5	82.8	77.9	61.8	43.2	30.9	(19)	
		MCWB	30.3	27.1	39.4	49.9	57.2	62.6	66.9	65.3	63.7	50.7	38.6	30.1	(20)	
	2%	DB	22.8	24.8	41.6	57.0	71.0	79.3	81.8	78.8	71.4	54.9	37.2	24.9	(21)	
		MCWB	21.9	22.9	35.0	45.0	53.1	59.4	63.5	62.9	60.2	47.7	34.7	24.1	(22)	
	5%	DB	18.4	18.5	35.9	50.8	65.3	76.0	79.1	75.0	65.7	49.7	33.7	20.7	(23)	
		MCWB	17.8	17.0	31.5	40.6	50.9	58.6	62.5	61.3	56.6	44.8	32.4	20.0	(24)	
	10%	DB	13.1	13.7	31.4	46.2	59.4	71.8	76.0	70.9	61.5	45.1	30.7	16.8	(25)	
		MCWB	12.5	12.3	28.6	37.9	48.1	56.8	61.3	59.6	54.7	41.7	29.6	16.3	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.0	27.6	39.8	50.8	58.8	68.0	70.6	67.9	65.5	52.2	38.8	29.9	(27)	
		MCDB	30.9	29.5	46.4	64.9	73.3	75.8	81.5	77.3	75.1	58.1	42.0	30.8	(28)	
	2%	WB	22.1	22.9	35.6	45.9	55.0	63.0	66.8	65.5	61.2	48.6	35.1	24.2	(29)	
		MCDB	22.8	24.5	40.8	55.9	67.2	74.5	76.8	74.3	68.7	55.2	36.9	24.9	(30)	
	5%	WB	17.8	17.0	32.2	41.8	51.9	60.4	64.7	63.3	58.3	44.8	32.4	20.1	(31)	
		MCDB	18.3	18.3	35.3	49.4	63.0	71.7	74.3	72.0	64.6	49.2	33.5	20.8	(32)	
	10%	WB	12.5	12.3	28.3	38.5	49.0	58.0	62.9	61.1	55.5	42.0	29.6	16.3	(33)	
		MCDB	13.1	13.6	31.4	45.5	58.8	69.1	72.2	68.2	61.1	45.1	30.7	16.7	(34)	
(36) Mean Daily Temperature Range	MDBR		16.3	21.7	25.3	23.9	21.7	22.8	21.2	18.6	16.3	11.8	13.8	14.5	(35)	
	5% DB	MCDBR	16.0	22.8	26.8	27.3	31.3	30.2	26.9	25.6	22.5	20.6	14.8	13.1	(36)	
		MCWBR	15.8	21.0	20.4	17.0	16.7	14.1	12.3	12.8	13.1	13.7	12.7	12.6	(37)	
	5% WB	MCDBR	15.7	22.4	23.3	25.3	27.8	24.5	22.4	20.6	19.8	19.2	14.0	13.0	(38)	
		MCWBR	15.6	20.7	18.0	16.2	15.9	13.3	11.6	12.1	12.6	13.3	12.2	12.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.241	0.239	0.272	0.330	0.336	0.357	0.402	0.356	0.324	0.287	0.251	0.215	(40)	
	taud		2.179	2.309	2.276	2.278	2.439	2.403	2.329	2.464	2.511	2.564	2.315	2.178	(41)	
	Ebn at Noon		215	272	286	282	287	281	264	272	267	254	214	199	(42)	
	Edh at Noon		17	23	31	36	33	35	36	30	25	18	15	13	(43)	
(44) All-Sky Solar Radiation	RadAvg		257	604	1111	1633	1773	1837	1698	1354	911	467	220	162	(44)	
	RadStd		24	45	86	132	145	163	112	117	103	66	22	23	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page